A Major Project Report
(Phase-III)

On

Sentinel-2 multispectral change detection for monitoring urban expansion and loss of water bodies

Submitted in partial fulfillment of the requirement for the award of the degree of

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1. Implementation Completed

During Week 3 of the project lifecycle, the team completed the full implementation of Project 26P_3CSE377, shifting from the prototype architecture to the complete geospatial change detection system. The implementation of all the modules has been completed from cloud raster ingestion through Google Earth Engine to local SQLite querying.

The main milestones in the implementation have been:

• Complete execution of the GEE pipeline for years 2018-2026: Seasonal Sentinel-2 Level-2A images have been processed for Greater Noida and Bengaluru Urban, hence generating the responsible flat files (area_stats.csv, area_stats_raw.csv, accuracy.csv, change_matrix.csv).

• Post-Classification Change Matrix Engine: Implemented 4x4 transition matrix calculations (T = M_2018 * 10 + M_2026) quantifying direct wetland loss, green cover conversion, and urban infill in square kilometers.

• Independent Ground-Truth Validation Framework: Digitized and verified 160 stratified validation points across Greater Noida and Bengaluru in Google Earth Pro (validation_points_google_earth.csv and validation_points.kml).

2. Code / Hardware / System Integration Status

The software stack is built in such a manner that allows easy and maintainable package management.

• pipeline/: Holds config.py, cloud_masking.py, spectral_indices.py, classifier.py, matrix_engine.py, sanity_checker.py, validation.py, and gee_pipeline.py.

• dashboard/: Holds data_loader.py, view_trends.py, view_maps.py, view_matrix.py, and view_accuracy.py doing the functionality of app.py.

• data/: Holds analysis tables (.csv), relational database (.sqlite), and overlays (.kml).

• tests/: Holds test_pipeline_and_dashboard.py doing 9 unit testing for schema verification, indexes calculation, temporal filtering, and accuracy calculations.

Current System Integration Status: Completed. The system works seamlessly in local mode (streamlit run app.py) and in production environment in Streamlit Community Cloud using CSV and SQLite data sources.

3. Testing, Validation & Performance Results

A rigorous verification process was carried out to verify the accuracy and scientific nature of the system:

• Automated Unit Testing: 9 unit tests were passed successfully (100% success rate in 0.12s using unittest/pytest).

• Step-2 Sanity Check: (1) checked that the CSV Schema has the correct number of lines (145, 145, 37, 33), (2) checked that the built-up area increases logically from 2018 to 2026 with no illogical reduction in values, (3) checked that the post-monsoon water value is more than pre-monsoon, and (4) checked that change matrix values are realistic (>98%).

Independent Ground-Truth Validation

The general accuracy of headline validation for the system is 89.38% with regard to 160 separate points of reference. This figure has been achieved by validating the system using 80 points from each city and 20 points from each class.

4. Current Milestone Completion Status

All of the predetermined milestones within the 3-week progression plan have been successfully reached:

• Milestone 1 (Week 1): Requirement gathering, Literature review, GEE authentication, and cloud masking pipeline creation - COMPLETED

• Milestone 2 (Week 2): Revision of v2 Random Forest classifier, 3-year majority filtering, non-reversion rule implementation, and Mid-Term assessment - COMPLETED

• Milestone 3 (Week 3): Google Earth independent validation, 4-view Streamlit dashboard deployment, unit testing, and final Phase III report compilation - COMPLETED

5. Pending Milestones / Remaining Work

All of the important technical and scholarly objectives of Project 26P_3CSE377 have been accomplished; the only outstanding tasks are those of completing the formatting of the PDF report and saving the project presentation, respectively.

6. Supporting Evidence / Repository / Demonstration Links

• GitHub Repository: https://github.com/devanshiSaini06/26P_3CSE377

• Supervisor Consultation Log: Weekly mentoring sessions conducted with Mr. Amit Agrawal (Associate Professor, School of CSE, IILM University, Greater Noida; email: amit.agrawal@iilm.edu) resulting in approval of the patched v2 temporal consistency framework.